

PROJECT VARDAN

Materials Intelligence, Climate Resilience & Fleet Logistics (LCA Stage A1–A4 & B1–B7)

1. Executive Summary

Project Vardan is an embodied carbon and climate resilience forecasting platform for the construction industry. Unlike conventional carbon calculators that only measure upfront factory manufacturing emissions (A1–A3), Vardan integrates **three synchronized modules** under the EN 15978 and ISO 21930 international standards:

- **LCA Module A1–A3 (Product Stage):** Baseline embodied carbon derived from the ICE V5 dataset across 259 building materials.
- **LCA Module A4 (Construction Stage - Transport):** Fleet logistics emissions factoring vehicle curb weight, payload mass multiplier, and transit distance.
- **LCA Module B1–B7 (Use Stage - Operational Degradation):** 100-year dynamic Global Warming Potential (GWP) simulated via a Random Forest Regressor under regional climate calamity stress.

2. Vehicle Fleet Database (LCA Stage A4)

The transportation engine supports four commercial logistics vehicle categories with calibrated baseline emission factors:

Vehicle Type	Curb Weight (kg)	Base Emission Rate	Fuel / Energy Type	Typical Role
Cargo Bike	120 kg	25.0 g CO2 / km	Human / Micro-e-cargo	Short-range urban micro-logistics
Electric Van	1,500 kg	40.0 g CO2 / km	Grid Electricity (Zero Tailpipe)	Clean urban light commercial
CNG Truck	970 kg	120.0 g CO2 / km	Compressed Natural Gas	Medium utility regional transport
Diesel Van	2,000 kg	180.0 g CO2 / km	Heavy-Duty Diesel	Standard industrial site haulage

3. Mathematical Logic & Physics Formulation

The transportation emissions calculation uses a continuous mass-penalty algorithm:

1. Gross Combined Weight: $W_{total} = W_{vehicle} + W_{load}$
2. Dynamic Load Multiplier: $\gamma = W_{total} / W_{vehicle} = 1 + (W_{load} / W_{vehicle})$
3. Total Journey Emissions: $Emissions_g = Base_Rate * Distance_km * \gamma$
4. Unit Conversion: $Emissions_kg = Emissions_g / 1000$
5. Normalized Carbon: $E_norm = Emissions_kg / W_{load}$ (in kg CO2e / kg material)
6. Lifecycle Total: $Total_Lifecycle = Base_GWP(A1-A3) + E_norm(A4) + Calamity_Penalty(B1-B7)$

4. Numerical Verification Example

Scenario: Transporting 1,000 kg of Blast Furnace Slag Cement over a distance of 50 km using a Diesel Van:

Step / Parameter	Formula / Source	Value	Unit
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Vehicle Curb Weight	Database (Diesel Van)	2,000.00	kg
Cargo Payload	User Input	1,000.00	kg
Gross Combined Weight	2,000 + 1,000	3,000.00	kg
Load Weight Factor	3,000 / 2,000	1.500	x multiplier
Transit Distance	User Input	50.00	km
Base Emission Rate	Database	180.00	g / km
Total Trip Emissions (g)	180 * 50 * 1.500	13,500.00	g CO2
Total Trip Emissions (kg)	13,500 / 1000	13.50	kg CO2
Normalized Material Transport (A4)	13.50 kg / 1,000 kg	0.0135	kg CO2e / kg
Baseline Embodied Carbon (A1–A3)	ICE V5 Database	0.2500	kg CO2e / kg
100-Year Dynamic Calamity GWP	ML Random Forest Model	0.2627	kg CO2e / kg
Net 100-Year Lifecycle Carbon	0.2627 + 0.0135	0.2762	kg CO2e / kg

5. API Endpoints Reference

- **GET /api/v1/materials:** Returns all 259 ICE V5 construction material names.
- **GET /api/v1/transport/vehicles:** Returns fleet vehicle catalog with weights and base emission rates.
- **POST /api/v1/transport/calculate:** Standalone calculation for journey transport emissions.
- **POST /api/v1/predict:** Unified prediction combining Stage A1–A3, A4, and 100-year B1–B7 dynamic resilience.